



Cloud Environments

Introduction to DevOps

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CONTENTS

- Introduction and background to cloud computing
- Cloud providers
- Cloud environments and web applications

Introduction

- In order to run software, a suitable runtime environment is needed
 - Software containerization is a way to package both the executable software and its runtime environment
 - However, the container itself isn't enough - it needs to be run in an environment where it is accessible to the target audience
 - Production environment means the setting where the latest working version of the software is run
- In this course, the aim is to publish a web software to everyone - therefore the production environment of the software should be accessible from the internet

Background information

- Traditionally, server computers have been used in order to deploy web software
 - The server computers might have resided physically in a company's premises
 - Another option has been to rent a dedicated / shared server from a provider
- There are some difficulties related to this approach:
 - It is hard to scale - especially the physical servers have a static amount of resources
 - Maintaining the environment (server backups etc.) isn't usually done by the provider
 - The servers need to be upgraded when time passes
- The idea of providing computing resources and services dynamically over the internet emerged

What is cloud?

- Cloud services have the following characteristics:
 - Scalability
 - Reliability
 - Security
 - Dynamic pricing
- In essence, cloud providers offer a variety of computing services that can be bought, accessed and used over the internet
-> this removes the difficulties presented in the earlier slide

Major cloud providers

- Amazon Web Services (AWS)
- Google Cloud Platform (GCP)
- Microsoft Azure
- There are also many other providers, like DigitalOcean, IBM, Alibaba Cloud etc.
- The major providers offer the widest range of selection regarding the cloud services
 - They also provide a CLI tool for experienced users
- Additionally, there are a lot of ready-made integrations between common software development tools and the major cloud services
 - For example, it is possible to integrate Visual Studio Code with Microsoft Azure via a VS Code extension
 - The integrations make the development work more efficient

Cloud environments and web software

- Web software (with or without containerization) requires a certain type of runtime environment
 - Java Spring application without containerization requires Java Runtime Environment
 - Node.js application without containerization requires Node runtime environment
 - All types of containerized applications require [containerd](#) as its runtime environment
- Cloud providers provide a variety of services that can be used to deploy web software - it requires knowledge to choose the most suitable service out of the options

Deploying web applications

- Let's consider the following situation
 - Software to be deployed: Java Spring Application
 - Cloud provider to be used: Microsoft Azure
- Suitable cloud services for deployment:
 - Azure Virtual Machines
 - Service which allows to provision Linux or Windows virtual machine
 - Azure App Service
 - Service which allows to run different kinds of applications (Java, Node.js, etc.)
 - The source code or the container image is provided to the service, which then runs the software automatically

Bonus: Infrastructure as Code (IaC)

- IaC means to manage and overlook the (cloud) infrastructure of a software product's runtime environment via certain types of definition files
 - Terraform is a good example of a IaC service: it is possible to manage cloud services for example in AWS, GCP or Azure with it
 - This automatizes the provisioning of cloud services and reduces the possibility for human errors (the definition files can be for example version controlled and assessed by everyone in the development team)
- > Terraform can be seen as an abstraction layer above the cloud services

THANK YOU!

